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Education

University of Delaware *Newark, DE, 19716, USA*
Graduate Student in Physics Sep. 2024 — Current

Xiamen University *No. 422, Siming South Road, Xiamen, Fujian, China*
B.S. in Chemistry Sep. 2020 — Jun. 2024

Professional Experiences

The University of Delaware *Newark, Delaware, U.S.*
Research Assistant Sep. 2024 — Current

- Advisor: *Dr. Tao E. Li*

The University of Oklahoma *Norman, Oklahoma, U.S.*
Research Assistant Aug. 2023 — Jan. 2024

- Advisor: *Dr. Yihan Shao*

Xiamen University & Fujian Chemistry Society *Xiamen, Fujian, China*
Teaching assistant & Assistant lecturer Sep. 2021 — Jun. 2024

- Teaching Assistant of Chemical Theory I and Mathematics in Natural Sciences I&II
- Assistant lecturer for Fujian provincial team of the 36th Chinese Chemistry Olympiad

Publications

(7) **Ji**, **X.**; Li, T. E. Nonlinear Freezing of Vibrational Polariton Transport via Mesoscale Simulations. **2026**, *Nano Lett. Accepted*, DOI: 10.48550/arXiv.2606.27463.

(6) Meng, G.; Vargas, A. F. B.; **Ji**, **X.**; Garcia-Gaitan, F.; Reyes-Osorio, F.; Varela-Manjarres, J.; Ren, Y.; Dinpajooh, M.; Nikolić, B. K.; Li, T. E. FermiLink: A Unified Agent Framework for Multidomain Autonomous Scientific Simulations. **2026**, DOI: 10.48550/arXiv.2604.03460.

(5) **Ji**, **X.**; Begušić, T.; Li, T. E. Two-Dimensional IR–Raman Spectroscopy of Vibrational Polaritons: Role of Dipole Surfaces. *J. Chem. Phys.* **2026**, *164*, 234116.

(4) **Ji**, **X.**; Bocanegra Vargas, A. F.; Meng, G.; Li, T. E. MaxwellLink: A Unified Framework for Self-Consistent Light–Matter Simulations. *J. Chem. Theory Comput.* **2026**, *22*, 2725–2738.

(3) **Ji**, **X.**; Li, T. E. Selective Excitation of IR-Inactive Modes via Vibrational Polaritons: Insights from Atomistic Simulations. *J. Phys. Chem. Lett.* **2025**, *16*, 5034–5042.

(2) **Ji**, **X.**; Pei, Z.; Huynh, K. N.; Yang, J.; Pan, X.; Wang, B.; Mao, Y.; Shao, Y. On the Entanglement of Chromophore and Solvent Orbitals. *J. Chem. Phys.* **2025**, *162*, 064107.

(1) Yan, S.; **Ji**, **X.**; Peng, W.; Wang, B. Evaluating the Transition State Stabilization/Destabilization Effects of the Electric Fields from Scaffold Residues by a QM/MM Approach. *The Journal of Physical Chemistry B* **2023**, *127 (19)*, 4245–4253.